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Research article

Trust gaps in energy transitions: Japan's National Deliberative Poll after Fukushima

Daphne Ngar-yin Mah [a,b], Darren Man-wai Cheung [a,b,*], Victor Wai Yin Lam [a,b,e], Alice Siu [c], Yasunori Sone [d], Ka-yan Li [a,b]

[a] Department of Geography, Hong Kong Baptist University, Hong Kong
[b] Asian Energy Studies Centre, Hong Kong Baptist University, Hong Kong
[c] Center for Deliberative Democracy, Stanford University, United States
[d] Professor Emeritus, Keio University, Japan
[e] Institute for Resources, Environment and Sustainability, University of British Columbia, Canada

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ABSTRACT

A global decline in public trust has created a strong need for governments worldwide to engage citizens in order to enhance policy legitimacy and ensure effective post-Fukushima energy transitions. Deliberative policy-making – an advanced form of participatory policy-making that emphasises dialogue and debate – is widely regarded as an intervention that can enhance trust and subsequently legitimacy of energy transitions policies. However, the potential and limits of deliberative policy-making remain the subject of debate. This paper contributes to this debate by exploring deliberative policy-making from a trust perspective. We develop a trust-based systems framework of deliberative policy-making in the energy transitions context to understand, examine, and conceptualise the quality of such policy-making processes. Our framework is tested and applied to a case study of a national deliberative poll (DP) on energy held in Japan in 2012. This study draws on qualitative and quantitative data derived from the DP, in particular from a two-day deliberative forum involving 285 citizens. The findings indicate that the existence of a trust gap may jeopardise the quality of deliberative processes where citizen participation is considered as an input and enhanced policy legitimacy as an output of a policy-making system. We also show that the trust gap is a complicated concept consisting of three dimensions (trust in information, motives, and competence) and three types of directional dynamics (vertical, horizontal, and temporal). Our study identifies the context of public distrust as well as the broader political environment as two critical contextual conditions that may inhibit the closure of the trust gap. Our study contains rich insights on deliberative policy-making in the energy context, arguing that it carries no guarantee of enhanced policy legitimacy. Policy-makers in the energy transitions field need to focus attention on creating the conditions to build public trust in order to enhance policy legitimacy and thus realise the potential benefits of deliberative practices of policy-making.

Abbreviations: DP, Deliberative poll; DPJ, The Democratic Party of Japan; EEC, The Energy and Environmental Committee; LDP, The Liberal Democratic Party.

* Corresponding author.
E-mail address: dmcheung@hkbu.edu.hk (D. M-w. Cheung).

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1. Introduction

Public trust is generally regarded as an essential component of effective government in all major areas of public policies. Trust can be defined as "a psychological state comprising the intention to accept vulnerability based upon positive expectations of the intentions or behavior of another" (Rousseau et al., 1998, p. 395). Trust has shown to be critical in enhancing policy legitimacy (Lövbrand et al., 2009; Motosu and Maruyama, 2016). Energy transitions have become a global trend amidst heightened concern about climate change. Many countries and cities are seeking to optimise their mix of energy technologies, from the use of nuclear power to the growth of renewable energy and the deployment of smart grids. Empirical evidence suggests that the demand for public trust is increasing in this context, given that there are many different routes to the desired end-state of low-carbon energy systems and the choices about these technological options have to be made on the basis of incomplete information. Furthermore, decision-makers must also factor in technological risks such as those associated with nuclear energy (Loorbach and Verbong, 2012; Mah et al., 2014).

Such an increase in demand is, however, accompanied by a global decline in public trust in reality (Gurría, 2013). The need to enhance public trust is even more pressing in the wake of the 2011 Fukushima nuclear accident in Japan, an event which exacerbated public distrust in governments, regulators, and energy industries not only in Japan, but around the world (Greenberg, 2013; He et al., 2014; Kinoshita, 2012). The widening trust gap has negative consequences, including deadlocks in energy policy-making (Chung and Kim, 2018; Mah and Hills, 2014; Mah et al., 2012; TUCK Foundation, 2018).

Deliberative participatory approaches to energy policy-making have attracted growing attention around the world as an innovative approach to enhancing trust, and thereby securing the policy legitimacy of post-Fukushima energy transitions. In contrast to traditional top-down and government- or expert-led public consultations, deliberative participatory approaches are commonly regarded as a more advanced way of securing public participation. Deliberative elements, such as stakeholder dialogues and public debates, are expected to enable the public to reach more informed views on complex energy matters based on trust relationships rather than blind acceptance (Abelson et al., 2003; McLellan et al., 2017; Petts, 2008; van de Kerkhof, 2006). Many countries, including Germany, France, the Netherlands, Japan, and South Korea have adopted deliberative approaches to policy-making in their search for energy transitions (CDD, 2012a; Kern and Smith, 2008; Public Deliberation Committee, 2017; Schneider, 2013; Schweizer et al., 2016). The citizen conference of energy transitions in France and the Public Deliberation on Shin-Gori nuclear reactors in South Korea are good examples, illustrating the diversity of deliberative policy-making practices (Gailing and Moss, 2016; Public Deliberation Committee, 2017).

However, deliberative approaches to policy-making cannot guarantee that trust will be built. Public trust may deteriorate further if people perceive that such deliberative engagement is not being properly conducted (Mah and Hills, 2014). The development of deliberative participatory approaches to policy-making has not been free from controversy, and debate continues on its potential and limits (Petts, 2008). In particular, there has been little research attention paid to the question of how deliberative policy-making processes determine consequences.

This paper aims to contribute to this debate by examining the relevance as well as the challenges of using a deliberative approach of policy-making to enhancing policy legitimacy of energy transitions. We develop a trust-based systems framework of deliberative policy-making for energy transitions, and test the framework in a case study of a national poll organised by the Japanese government in 2012. By considering this case study, we aim to offer a better understanding of deliberative policy-making process from a trust perspective. Our systems perspective argues that a good understanding of deliberative energy policy-making must look beyond outputs and give sufficient attention to the process (Schmidt, 2013). This study set out to answer the following research questions:

- How can the trust gap be measured?
- Whether the deliberative process facilitate or impede the closure of the trust gap; and if so, what were the outputs in terms of energy transitions legitimacy?
- What were the key factors facilitating or constraining the closure of the trust gap by using the deliberative process?

The 2012 National Deliberative Poll on Energy and Environmental Options in Japan ("the 2012 national DP") was conducted by the Japanese government slightly more than a year after the 2011 Fukushima nuclear accident. Approximately 7000 sampled registered voters participated in a telephone poll in July 2012, and 285 subsequently joined a two-day deliberative forum in August in which they were asked to vote on three scenarios: 0% nuclear, 15 % nuclear and 20−25% nuclear energy in the electricity mix in 2030 (White, 2015).

The 2012 national DP merits study for two reasons. Firstly, it enables one of the few rigorous examinations of whether deliberative participatory processes can enhance trust, and in doing so improve policy legitimacy. Such a focus on trust may seem unfair, since the Japanese administration has never explicitly stated that this was the purpose of the 2012 national DP. Instead, its focus was to enhance the legitimacy of its post-Fukushima energy strategies. Before the 2012 national DP took place, the Japanese government pledged that it would consider citizens' recommendations to formulate a national energy strategy in August 2012 – one month after the participatory event (Transcript A002). In light of the strong need for governments around the world to enhance public trust in relation to energy transitions plans, our case study of Japan will make contribution to a better understanding of the relevance of trust to transitions legitimacy. Secondly, the 2012 national DP is one of the very few national-scale energy-related deliberative participatory exercises that have been carried out worldwide. Accordingly, it provides a wealth of accessible empirical data, presenting an invaluable opportunity to analyse the deliberative process in depth.

The paper is structured as follows. Section 2 provides a theoretical discussion. Section 3 discusses our methodological approaches. Section 4 presents the case contexts. Section 5 presents our analysis. The last section offers some concluding thoughts and policy

Table 1
A review of the three dimensions of trust and their associated indicators.

Trust dimensions	Trust indicators	Brief illustrations	Sources
Trust in information	Transparency	The government provides information proactively in a meaningful, accessible form free of charge or at a reasonable cost	(Irwin, 2006; Mah and Hills, 2014)
	Comprehensiveness	The government provides all relevant information about the subject matters	(Rowe and Frewer, 2000; Walker et al., 2008)
	Objectivity and reliability	The government provides participating parties balanced information that includes a variety of perspectives rather than biased or partial information, or misinterpretation of information	(Mah and Hills, 2014; Rowe and Frewer, 2000)
Trust in motives	Openness	Opening-up process rather than pre-empting the results of consultation or participatory exercises	(Stirling, 2008)
	Integrity	The government is seen as honest, or not hiding information about risks and manipulated a participatory event for her own sakes. The government is not too influenced by business	(Kim, 2005; Walker et al., 2008)
	Inclusiveness	The government listens to opposing arguments	(Rowe and Frewer, 2000; Schweizer et al., 2016)
	Representativeness	Citizens of different background and viewpoints are invited to participate in deliberation so that such participation is representative of the needs and demands of the public	(Redburn et al., 1980; Renn et al., 1993)
	Credible political commitment	The government actor has interest and determination to honour her agreements and is consistent in its political decisions	(Kim, 2005)
	Perceived policy outcomes	Whether participants perceive that their considered views could make an impact on policy	(Mah and Hills, 2014)
Trust in competence	Capability in effective operation	The government is seen as acquiring the necessary knowledge and skills for effective operations	(Kim, 2005)
	Capability in risk management	The government is perceived to be capable of dealing with environmental or energy-related risks (including nuclear risks)	(Walker et al., 2008)

recommendations.

2. The theoretical perspectives of energy transitions, trust, and deliberative policy-making

2.1. Trust and the nature of energy transitions

Global climate change, energy security and post-Fukushima nuclear concerns have heightened the urgency of sustainable energy transitions around the world. Such transitions generally involve a shift from centralised, fossil fuel or nuclear-based energy production systems (Loorbach and Verbong, 2012) to more decarbonised, decentralised, and inclusive ones (Winskel, 2018). Sociotechnical energy transitions theory, with its roots in technological innovation systems studies, demonstrates that such a structural transformation needs complex, systemic changes in markets, institutions, policies, technologies, behaviour, and culture (Geels, 2002; Verbong and Loorbach, 2012).

The challenges of governing such complex changes in energy systems have drawn attention to the concept of trust. A growing body of the energy studies argues that trust is a predictor of public acceptance, which itself is one of the critical dimensions of policy legitimacy (Lövbrand et al., 2009; Motosu and Maruyama, 2016). Legitimacy, which generally refers to the acceptance and justification of a shared rule by a community, requires political decisions to be derived, directly or indirectly, from citizens' preferences through transparent, inclusive, and accountable policy-making processes (Lövbrand et al., 2009; Turner et al., 2016). As such, trust can be regarded as a social licence to operate (Smits et al., 2017). The more people trust energy regulators and other responsible parties, the more likely they would voluntarily cooperate with regulators, support energy plans or participate in energy projects even when these go against their own interests or where they have insufficient information (Mah and Hills, 2014; Mah et al., 2014). This is because people will assume that the decisions made by the government in putting such plans into place are legitimate and will protect their interests (Kalkbrenner and Roosen, 2016; Kim, 2005).

Empirical studies of energy transitions suggest that there is an increase in the demand for trust in this context. Energy transitions have a special kind of trust requirement because they have inherent uncertainties and risks. The literature emphasises that there is no single known pathway to transitions, but several possibilities (Bellaby, 2007). Public trust in various parties (including governments, experts, media, and NGOs), institutions, and organisations is critical in the context of energy transitions because many decisions about energy choices must be made on the basis of incomplete information (Mah and Hills, 2014) while many technological risks (including nuclear risks) are often value-laden (Loorbach and Verbong, 2012; Mah and Hills, 2014).

However, the decline in public trust has become a global phenomenon. Empirical studies have documented that citizens lose trust in political decisions and policy-makers due to heightened concerns about regulatory incompetence, limited enforcement, and conflict of interest in dealing with controversial energy policies (Greenberg, 2013; Mah et al., 2014). The disconnection between societal demands for trust and the perceived trustworthiness of governing institutions have negative consequences. The large-scale social movements in recent years, including the Yellow Vest Movement that started in late 2018 in France and the global climate strikes in 2019, show that governments and institutions are widely seen as falling short of fulfilling the expectations of trust in their practices around energy decision-making (Huvos, 2019; Kipfer, 2019).

2.2. The trust gap as a concept

Although the energy transitions literature does not discuss the trust gap explicitly, the concept has been relatively well developed in the political, business, and environmental fields. Scholars in these studies define the trust gap as the difference between the trust needed by stakeholders and the trust possessed by governments, institutions, and corporations (Pirson, 2009). The trust gap can occur when a trustor desires a greater demonstration of trust than what a trustee is willing or able to provide (Lacey et al., 2018).

There is a rich literature on trust which sheds light on how to conceptualise the trust gap from various aspects. Firstly, trust has several key dimensions within which the gap may lie. These include trust in information (Frewer et al., 1996), trust in motives (Coleman, 1990; Hardin, 1996; Twyman et al., 2008), and trust in competence (Mayer et al., 1995; Twyman et al., 2008). The literature suggests that each dimension has its own set of indicators. ***Trust in information*** is associated with: (i) transparency (Frewer et al., 1996); (ii) comprehensiveness (Rowe and Frewer, 2000); and (iii) objectivity and reliability (Rowe and Frewer, 2000). ***Trust in motives*** is associated with: (i) openness (opening or closing down processes) (Stirling, 2008); (ii) integrity (Kim, 2005; Walker et al., 2008); (iii) inclusiveness (Rowe and Frewer, 2000; Schweizer et al., 2016); (iv) representativeness (Redburn et al., 1980; Renn et al., 1993); (v) credibility of political commitments (Kim, 2005); and (vi) perceived policy impact (Mah and Hills, 2014). ***Trust in competence*** is associated with: (i) operational capability (Kim, 2005); and (ii) capability in managing energy-related risks (Walker et al., 2008). Table 1 provides an overview of these three dimensions of trust, their indicators, and brief illustrations of each.

The trust literature also makes a directional distinction in terms of relationships with different parties. Vertical trust refers to relationships between leadership or government and citizens, whereas lateral (or horizontal) trust refers to relationships among peers, individuals, and organisations in a society who are in similar situations (Guinot and Chiva, 2019). The temporal dimension of trust underscores the importance of time, with pre-existing trust as one of the contextual factors critical to the effectiveness of participatory policy-making (Dawson and Darst, 2006). Furthermore, political context is another feature identified in the literature as critical in facilitating or constraining the pressure on the creation, development, and maintenance of trustworthy relationships, or bridging the trust gap (Khodyakov, 2007; Sztompka et al., 2000).

2.3. Deliberative policy-making as an approach to building trust

How, then, can public trust be enhanced? Deliberative policy-making is one of several participatory practices that have been increasingly advocated and adopted by governments around the world to build trust and enhance policy legitimacy (Mah and Hills, 2014; Mah et al., 2014; Travaline et al., 2015). In contrast to traditional forms of policy-making that mostly take place in top-down bureaucratic structures, deliberative policy-making is a participatory policy process that seeks to improve the quality of policy-making by involving citizens (Bua and Escobar, 2018; Travaline et al., 2015).

Deliberative processes emphasise dialogue, mutual learning, and debate in which participating citizens can exchange viewpoints, weigh and balance arguments, and offer reflections (Eun, 2016; van de Kerkhof, 2006). The normative outcomes of deliberative participatory processes include the enhancement of public trust and therefore greater policy legitimacy (Bull et al., 2008; Petts, 2003). Many countries, including Germany, France, the Netherlands, Japan, and South Korea have engaged in deliberative policy-making in the search for energy transitions (CDD, 2012a; Kern and Smith, 2008; Public Deliberation Committee, 2017; Schneider, 2013; Schweizer et al., 2016). The 2012 national DP is a good example, where selected citizens were invited to envision and deliberate their energy futures.

Nevertheless, debate continues on the potential and limits of deliberative policy-making. Although there has been an extensive academic exploration of the significance of deliberative democracy and participation in the broad fields of political studies and environmental governance (e.g. Drazkiewicz et al., 2015; Habermas, 2006; Newig, 2007; Newig and Fritsch, 2009; Welsh, 2002), as well as in the specific field of transitions management (e.g. Avelino et al., 2016; Hendriks, 2009; Voß et al., 2009), the empirical evidence on the outcomes of deliberative approaches is mixed. Some studies argue that deliberative practices cannot guarantee positive outcomes. Public trust can be even further eroded if the public perceives procedural biases in the deliberative process (Mah and Hills, 2014), whilst political factors, particularly the uneven distribution of power in society, may limit the extent to which the normative benefits can be realised (Bloomfield et al., 2001).

2.4. A systems perspective of deliberative energy policy-making

As a subfield of political science, systems approaches offer a potentially useful way of understanding deliberative energy policy-making. Based on the work of Easton (1957), systems scholars conceptualise policy-making systems, as well as other types of systems such as governance systems and technological innovation systems, as a unit (Stewart and Ayres, 2001). Systems studies suggest that citizen participation can be defined as inputs into systems, which is then converted into outputs through the process of deliberative interactions within the political system (Schmidt, 2013). Citizen input in support for deliberative policy-making can occur in different forms which include direct elections for government, citizen juries, and deliberative polls (Schmidt, 2013). Outputs are political decisions or policies; these have consequences which may include enhanced trust and legitimacy (Schmidt, 2013).

Systems scholars have paid increasing attention to processes, or the "throughput" dimension - the missing element between inputs and outputs, which has hitherto been under-studied. Process-oriented systems studies argue that the quality of the *governing process*, and not only the effectiveness of the outcomes and the participation of the citizenry (the input), is an important criterion for evaluating a system's overall performance towards achieving desirable states such as legitimacy (Lieberherr and Thomann, 2020; Schmidt and Wood, 2019; Schmidt, 2013). This process-oriented focus is particularly useful because deliberative processes cannot guarantee that good outputs will be produced. The literature suggests that bad processes, involving incompetence, biased governance practices, or procedural inequity can undermine legitimacy (Schmidt, 2013).

Despite the calls to refocus attention on the deliberative process and the increasing policy interest in public distrust of energy transitions decisions, what is missing from such a systems perspective is a conceptualisation of a trust-based deliberative participatory mechanism. This is particularly so in the Asian context. This case study sets out to fill these gaps, and is therefore of scholarly importance.

2.5. Towards an integrated trust-based systems framework for deliberative energy policy-making

In this paper, we present a new integrated trust-based systems framework for deliberative energy policy-making (Fig. 1). It integrates the sociotechnical transitions (Geels, 2002; Verbong and Loorbach, 2012), trust (Frewer et al., 1996; Smits et al., 2017), deliberative policy-making (Bull et al., 2008; Meadowcroft, 2004), and systems perspectives (Schmidt, 2013). The framework focuses on the quality of the deliberative process, and offers a conceptualisation of deliberative policy-making processes in the context of energy transitions, with a focus on the trust perspective. Based on insights drawn from the trust gap literature (Hill and Donaldson, 2015; Lampton, 2017), this study defines the trust gap as the difference between the desirable and optimal levels of public trust and the perceived trustworthiness of governments and other trustees (most notably regulators and experts) in the context of energy transitions. Our systems framework draws attention to the processes which can close the trust gap with a view to guiding the analysis of how deliberative policy-making processes can facilitate or impede such closure.

Three propositions can be derived from our framework. Firstly, citizen input to a policy-making system is a type of policy intervention in response to demand for public trust in the context of energy transitions. Such input can be converted through deliberative processes into normative outputs which include enhanced trust and, through that, increased policy legitimacy. Secondly, trust in information, motives, and competence are the three key dimensions of the trust gap that form the key evaluative aspects of the quality of deliberative processes. Thirdly, contextual factors, including distrust context and political conditions, will impede the trust gap from closing.

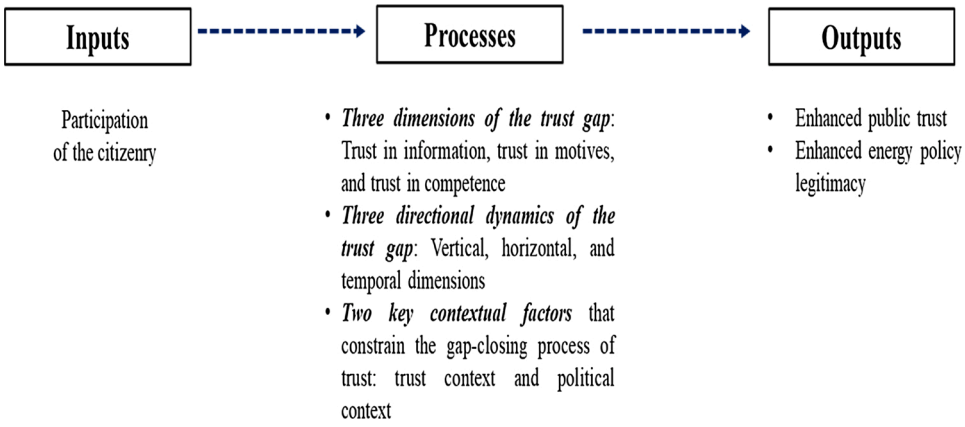

Fig. 1. A trust-based systems framework for deliberative policy-making in energy transitions.

3. Research approach and methods

This research is a single case study that adopts a multimethod approach. Using a case study approach enables us to understand and examine the complex dynamics and process of the 2012 national DP (Eisenhardt, 1989). It draws on four main sources of information: (i) a qualitative document analysis; (ii) a secondary analysis of published quantitative data generated by three sets of questionnaires (a telephone opinion poll (n = 6849), a pre-forum questionnaire (n = 285), and a post-forum questionnaire (n = 285)(CDD, 2012c));[1] (iii) a primary qualitative analysis consisting of approximately 70 hours of video footage recorded at 40 small-group discussions and two plenary sessions; and (iv) the direct observations conducted by two of the authors who had direct involvement in the design and implementation of the 2012 national DP.

A major data source is the qualitative data derived from the transcriptions of the 2012 national DP. A total of 40 small-group discussion sessions (two for each of the 20 small-groups convened) and two plenary sessions (where the 20 small-groups had the opportunity to meet and discuss) were conducted at the 2012 national DP; all were video-recorded. A total of approximately 70 hours of footage was transcribed into Japanese (732 pages) and then translated into English (639 pages) by research assistants. This generated a wealth of primary data. While we conduct a secondary analysis of the quantitative data here, this study makes an important contribution to the field through production and consideration of the primary qualitative data described above. Based on our framework, the transcribed data is coded for theme analysis.

The use of multiple datasets improves the quality of our analysis in two important ways. Firstly, it can enable us to derive combined insights on the phenomena observed. Secondly, we can enhance the robustness of our analysis through the triangulation of different methods (questionnaires and small-group discussions) and datasets (LeCompte and Goetz, 1982; Miles et al., 2014; Tobin and Begley, 2004).

4. Case contexts

4.1. The national and energy context of post-Fukushima Japan

Japan is a major developed and industrialised OECD country, with a population of about 126 million in 2019. Ranking a global third in terms of GDP after the US and China (IMF, 2019; Statistics Bureau, 2020), Japan is heavily reliant on imports to meet about 90 % of its energy needs (WNA, 2019) and is the world's fourth largest electricity consumer (CIA, 2019). As early as the 1970s, driven by the problems of excessive oil reliance as revealed in the Oil Crises, the Japanese government had already started to proactively promote nuclear power and energy conservation (CDD, 2012a). In 2010, approximately 60 % of Japan's electricity was generated from fossil fuels and 31 % from nuclear energy (ANRE, 2019).

The Fukushima nuclear accident took place on 11th March 2011 and marked a watershed in Japan's energy policy. There was a drastic shift in nuclear-related policies. Several months before the accident, the 2010 Basic Energy Plan had set out to double the share of nuclear-based electricity to 50 % by 2030. This was brought to a close immediately after the accident, with all 54 operating nuclear reactors[2] in Japan shut down for review in March 2011, meaning that 0% of Japan's energy was produced by nuclear. In September 2012, the Energy and Environmental Committee (EEC) released a policy entitled "Innovative Strategy for Energy and the Environment" which proposed to phase-out nuclear energy completely by 2030.

However, nuclear restart programmes gradually accelerated following the change of political leadership from the anti-nuclear

[1] These primary quantitative data are available from the website of the Center of Deliberative Democracy, Stanford University (CDD, 2012c). Appendix A provides the pre- and post-forum questionnaires.

[2] There were 57 reactors in total and three reactors under construction (nippon.com, 2019; Srinivasan and Gopi Rethinaraj, 2013).

Democratic Party of Japan (DPJ) to the pro-nuclear Liberal Democratic Party (LDP) in December 2012. Accordingly, the proportion of Japan's total electricity generated from nuclear sources rose from approximately 1.9 % in 2016 to 7.1 % in 2019 (ANRE, 2019; WNA, 2019) (Fig. 2). As of December 2019, nine reactors had resumed operation and 24 had been permanently shut down (JAPC, 2019). Appendix B provides an overview of the changes in political regimes, nuclear strategies, and nuclear policies in Japan between 2010 and 2018.

In terms of the political context, Japan is one of the few Asian countries to have a relatively long tradition of participatory policy-making. In Japan's high-level politics, the government institutions and policy-making processes tend to be top-down (Boswell et al., 2013). But as early as the first half of the 1990s, the Japanese government started to introduce major initiatives to involve citizens in policy-making through a wide variety of participatory approaches. The formats of such initiatives range from public hearings to roundtable conferences, referendums, consensus conferences, deliberative polls, planning cells, and town hall meetings (Appendix C). It is important to note that while the 2012 national DP was a relatively innovative and sophisticated methodology for engaging the public in terms of Asia as a whole, it was not the first time Japan had used this methodology. Six other DPs on non-energy related issues had been conducted between 2009 and 2014.

4.2. The 2012 national DP

The 2012 national DP was a government-led national event. 285 citizens, selected from a larger random sample of 6,849 registered voters that participated in a preliminary telephone poll (T1) in July 2012, were recruited to participate in a two-day deliberative forum in August. In this forum, participants engaged in a national debate on three energy options for Japan: 0% nuclear, 15 % nuclear and 20−25% nuclear energy in electricity mix in 2030 (White, 2015), and were asked to fill in a pre-forum questionnaire upon their arrival (T2), and then a post-forum questionnaire by the end of the DP event (T3). This national debate was an official government-led public participation process that adopts the trademarked techniques of the Deliberative Poll ® (CDD, 2012a; White, 2015).

The methodology of scenario development was adopted to facilitate deliberation. Three energy scenarios were developed to facilitate participants' understanding and allow them to compare the benefits, weaknesses and risks of three alternative energy pathways. Iterative processes, comprising two rounds of small-group discussions (20 focus groups) and two plenary sessions, were facilitated to promote dialogic interactions among participants. Participants were also offered the opportunity to meet, raise questions with, receive feedback and clarify issues from a panel of experts at the plenary sessions, and subsequently made considered views on the energy options for the country.

A timeline of the key activities of the 2012 national DP and the policy impacts is provided in Appendix D.

5. Analysis

As an example of utilising our framework, we examine the 2012 national DP in Japan through the lens of a trust-based systems perspective. We found that when measured against the perspective of trust, the deliberative policy-making process did not deliver its normative benefits fully – at least as practised in the energy transitions in this case study of Japan. The existence of the trust gap appeared to jeopardise the participatory quality of the deliberative policy-making process. Guided by our framework, we have four key findings in relation to the trust gap and the deliberative process.

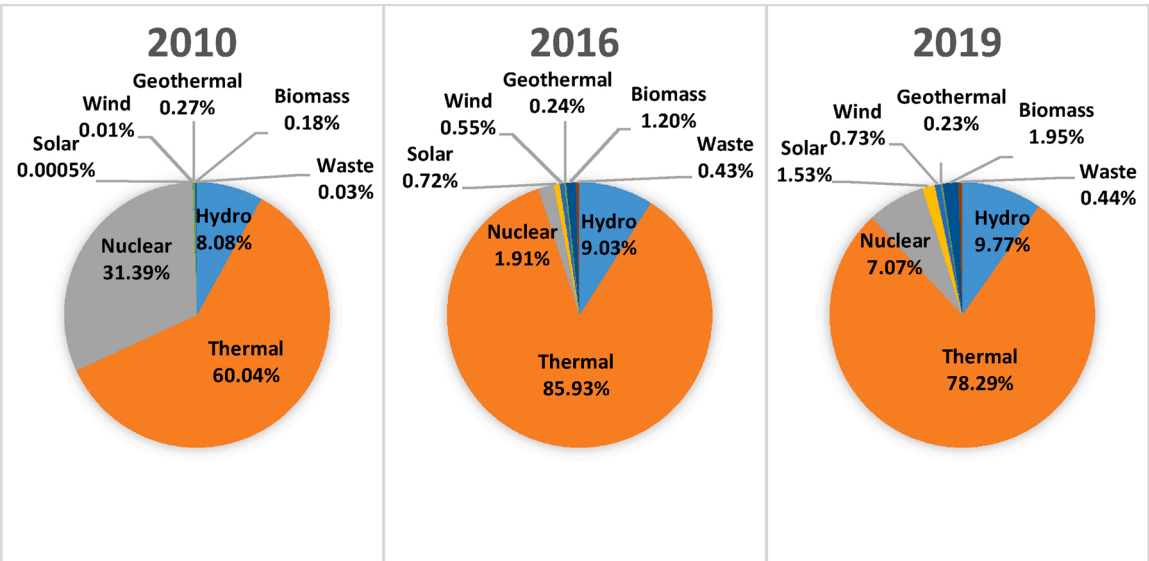

Fig. 2. Japan's electricity mix in 2010, 2016, and 2019.
Source: Authors; data compiled from ANRE (2019).

5.1. Three critical dimensions of the trust gap in deliberative processes

In order to understand the nature of the trust gap, one important question needs to be answered: Which dimensions or types of (dis) trust matter, and in what ways? This study uses qualitative data derived from the transcripts of the 40 small-group discussion sessions and two plenary sessions to complement the quantitative data to partially answer this question.

We found that (dis)trust in the three key dimensions, (dis)trust in information, (dis)trust in motives, and (dis)trust in competence were found to be critical in the deliberative process. We also derived a set of illustrative examples to shed light on the specific trust-conducive indicators in deliberative processes in Appendix E.

5.1.1. (Dis)trust in information

Our data suggest that trust in information mattered in the 2012 national DP. Participants' concerns were related to the three key aspects of information: transparency, comprehensiveness, and objectivity and reliability of the materials and expert views presented in the 42-page briefing document and during the deliberative forum. The participants were concerned about the following aspects:

(1) *Transparency:* The participants were concerned about *transparency* of the information (e.g. a participant was concerned that the poll organiser was not open enough to disclose the demographic characteristics of participants (Illustrative Example No. 1);
(2) *Comprehensiveness*: Participants raised concerns that there was: (i) a lack of non-government sources of information (Illustration Example No. 2); (ii) a lack of detailed information (Illustrative Examples No. 3, 4, & 9); (iii) a lack of explanation and elaboration (Illustrative Examples No. 5 & 8); and (iv) a lack of information so that they could not compare the benefits and weaknesses of alternative scenarios (Illustrative Example No. 7). Some participants noted that such limitations in information constrained them from engaging in meaningful discussion and they could not proceed further in discussion (Illustrative Examples No. 9 & 35);
(3) *Objectivity and reliability*: Some participants raised concerns over the *objectivity* and *reliability* of information. Some participants attempted to counter-check whether the information provided in the briefing document was non-biased (Illustrative Examples No. 11 & 12) or evidence-based (Illustrative Example No. 10).

5.1.2. (Dis)trust in motives

Our data show that public trust in motives also mattered in the 2012 national DP. Participants seriously questioned the motives of the government for six main reasons:

(1) *Openness*: Rather than an opening up process which could have invited public inputs to explore alternative energy transitions scenarios, some participants felt that the DP was designed in ways to force them or mislead them to choose the "middle" option, i.e. the 15 % nuclear option which was the government's preferred option. Some participants felt that the design of the three scenarios was a closing down process rather than an opening up one (Illustrative Examples No. 13-19);
(2) *Integrity*: It is evident that the government and power companies were perceived to be dishonest by some participants because of the negative experiences associated with a scandal of a number of post-Fukushima energy public hearings which took place before the 2012 national DP. Employers of power companies were alleged to have been selected to express pro-nuclear views as participants in those public hearings (Sieg, 2012). The government and power companies were alleged to manipulate the deliberation of those public hearings, and this resulted in some participants having negative expectations about the intentions of the government in organising the DP (Illustrative Examples No. 20-24). Additionally, some participants questioned whether the government might misuse or even manipulate the polling results for its own advantage (Illustrative Examples No. 25 & 26);
(3) *Inclusiveness*: Quantitative data suggest that participants had divided views on whether the information provided in the briefing document was inclusive enough to include opposing views. Although 34 % of participants agreed that the briefing document *included different opinions equally,* 26 % *of them* disagreed with that and 36 % were in the middle" (CDD, 2012c). Qualitative data also indicated that some participants were concerned that the briefing document did not cover different viewpoints on a broad range of alternatives. Some commented that, for example, certain types of renewable energy technologies such as wave power were not included in the briefing document (Illustrative Example No. 28);
(4) *Representativeness:* We found that the representativeness of the participating citizens matters. Rather than having trust in representativeness, the participants perceived that the DP was over-represented by anti-nuclear participants (Illustrative Examples No. 29-31);
(5) *Credibility of political commitments*: Some participants questioned the political commitment as there was an imminent regime change from the DPJ-led government to a LDP-led government in the national election by the end of 2012, and they were concerned that the Japanese government would not be able to maintain policy consistency due to political regime change (Illustrative Examples No. 32 & 33);
(6) *Perceived policy impacts*: Some participants perceived that the result of the DP would not impact on energy policy-making (Illustrative Examples No. 34-40). They were concerned that the decisions made in the DP would be ignored, and would not be implemented as an energy policy.

5.1.3. (Dis)trust in competence

Our data suggest that trust in competence mattered in the 2012 national DP. Participants' concerns related to two aspects of competence were as follows:

(1) *Perception of the government's implementation capabilities*: Some participants were concerned whether the government would be capable of implementing its energy decisions even such decisions were made. They questioned whether the DPJ government could implement the zero-nuclear option by the planned timeline, which was August 2012 (Illustrative Examples No. 41 & 42).
(2) *Perception of the government's competence in managing risks:* The experience of the 2012 Fukushima accident has also made some participants highly skeptical of the government's capacity to deal with nuclear risks and implement effective contingency measures when major nuclear accidents occur (Illustrative Examples No. 43-45).

5.2. Three types of directional dynamics of the trust gap

By combining quantitative and qualitative data that we derived from the 2012 national DP, our analysis can make a directional distinction on the trust relationships across different parties. We can specify the vertical, horizontal, and temporal dimensions of the trust gap that can help explain the complexity of the trust gap as a concept.

In the 2012 national DP, participants were asked in their questionnaires to evaluate the trustworthiness of different information sources in three points of time, i.e. T1, T2, and T3. Our secondary analysis of the published quantitative data (CDD, 2012c) shows that an extremely low level of *pre-existing* trust prevailed *after the DP*. Before the DP, at T1 (when participants were first contacted by the project team by email or phone in July 2012 – approximately one month before the DP forum was held), only 6.4 % of participants trusted information from the government, 3.6 % trusted information from electricity companies, 19 % trusted information from nuclear power specialists, and 11.7 % trusted information from the media (CDD, 2012c). The data suggest that the DP was conducted in a highly distrusting context.

After deliberation, there remained a consistently low level of trust towards all information sources available to the participants as follow:

(1) **Government:** "Distrust or do not trust at all" (0–4) *increased* from 67.4 % (T2) to 69.8 % (T3); "in between" (5) decreased from 26.3 % (T2) to 23.5 % (T3); and "trust or trust very much" (6–10) slightly increased from 6.1 % (T2) to 6.5 % (T3);
(2) **Nuclear power specialists*: "**distrust or do not trust at all" (0–4) *decreased* from 48.5 % (T2) to 42.8 % (T3); "in between" (5) increased from 31.9 % (T2) to 34.7 % (T3); and "trust or trust very much" (6–10) increased from 18.3 % (T2) to 21.4 % (T3);
(3) **Electricity companies: "**distrust or do not trust at all" (0–4): *slightly decreased* from 83.1 % (T2) to 81.4 % (T3); "in between" (5) slightly increased from 12.3 % (T2) to 13.7 (T3); and "trust or trust very much" (6–10) remain unchanged (4.0 %) (T2 to T3);
(4) **Media: "**distrust or do not trust at all" (0–4): *slightly increased* from 56.1 % (T2) to 57.1 % (T3); "in between" (5) decreased from 36.5 % (T2) to 34 % (T3); and "trust or trust very much" (6–10) slightly increased from 6.7 % (T2) to 7.8 % (T3);
(5) **Non-profit organisations (NPOs) or non-government organisations (NGOs)*: "**distrust or do not trust all" (0–4): *decreased* from 26.3 % (T2) to 23.6 % (T3); "in between"(5) decreased from 46 % (T2) to 40 % (T3); and "trust or trust very much"(6–10) increased from 23 % (T2) to 31.9 % (T3); and
(6) **Internet:** *"distrust or do not trust at all" (0–4): decreased* from 38.3 % (T2) to 35.4 % (T3); "in between" (5) slightly decreased from 44.9 % (T2) to 43.5 % (T3); and "trust or trust very much" increased from 11.6 % (T2) to 16.6 % (T3)

*statistically significant ($p < 0.05$) (CDD, 2012c).

It is important to note several observations for the trust gap as follows:

(i) The *vertical* dimension of the trust gap: Although this case study is not intended to provide a robust quantitative analysis of the trust gap, the consistently low scores of trust assigned to all information sources across T1, T2 and T3 indicate that the vertical gap, i.e. the discrepancy between the citizens' expectation, and the perceived trustworthiness of these trustees was considerable. In other words, these trustees are seen as falling short of fulfilling the trust expectation as perceived by the citizens.
(ii) The *horizontal* dimension of the trust gap: In the low trust context, it is interesting to note that horizontally there existed discrepancy of the (dis)trust levels across different parties or trustees. Government, electricity companies, and media received the lowest level of public trust (with respondents who "trust or trust very much", government, electricity companies and media amounted to 6.5 %, 4.0 % and 7.8 % respectively in T3) while NPOs appeared to receive the highest level of public trust (with respondents who "trust or trust very much" NPOs reached 31.9 % in T3).
(iii) The *temporal* dimension of the trust gap: Our case study found that low levels of trust prevailed both before (T1 and T2) and after deliberation (T3). This finding suggests that distrust was relevant not only as an outcome of the DP, but also as a context before the DP took place.

5.3. Mixed outputs in enhanced trust and policy legitimacy

What, then, were the outputs of deliberative policy-making processes? Was trust and policy legitimacy enhanced through deliberative processes? Our case study can shed light on the complexity of the relationships as follows:

Firstly, our case study found that the DP did lead to some, though modest, positive outputs in terms of enhanced trust and policy legitimacy. Even though the DP was in a largely distrust context, of particular interest is that the deliberative processes further reduced trust in government and the media whilst enhancing trust on four other parties as information sources: nuclear power specialists,

electricity companies, NGOs, and Internet between T2 and T3 with statistical significance (CDD, 2012b). The changes in participants' responses from T2 to T3 therefore provided evidence to show the deliberative processes changed their trust levels on various information sources. But the DP as a single deliberative event was far from enough to bridge the substantial trust gap as the overall trust levels on all trustees remained consistently low even after deliberation (T3).

Secondly, the DP appeared to influence policy legitimacy of a radical energy transitions policy, i.e. a nuclear phase-out policy decision, in a highly complicated way. A main result of the DP was the citizen recommendation on a "0% nuclear" option. The quantitative data from the poll suggest the "0% nuclear" option and support for no nuclear increase after deliberation were most preferred by the 285 participants. Increase in support for "0% nuclear" was observed from 60.5 % at pre-forum (T2) to 67.4 % at post-forum (T3). In contrast, decrease in support for "15 % nuclear" and "20−25% nuclear" was observed from 41.4 % and 24.3%–40.4% and 23.9 % respectively (CDD, 2012c). One month after the completion of the DP, in September 2012, the Japanese government recommended a nuclear phase-out plan, pledging to achieve a 0% nuclear target by 2030 (EEC, 2012). This political decision appeared to be in line with the DPJ government's aim of organising the DP – to collect citizens' considered opinions and consider the citizens' recommendations to formulate a national strategy (Transcript A002).

It is however important to note that the political decision of a nuclear phase-out policy was a short-lived one that lasted only for several months. The Liberal Democratic Party (LDP), a pro-nuclear party that won the national election in December 2012 did not inherit the 0% nuclear plan. Since then, Japan has gradually restarted the use of nuclear power. Nuclear restart applications began to be reviewed in mid-2013, and the 4th Basic Energy Plan which resumed the nuclear option was introduced in 2014 (Appendix B).

5.4. Distrust and political factors constrained deliberative energy policy-making processes

Whilst we would not expect one single DP to be able to restore public trust in Japan, it is still important to examine why, despite the efforts made at the national level to engage the public to deliberate on energy futures, a low level of trust remained after the DP. Our study found that the distrust and political context of the DP appeared to create inhibitive settings in which deliberative processes cannot realise its full normative potentials.

In relation to the distrust context, this DP was organised at a time when public distrust in government, regulators, and nuclear regulators in Japan was heightened by the Fukushima accident. In early July 2012, just a few weeks before the event day of the DP, the official report conducted by the National Diet of Japan was released. It stated that the Fukushima nuclear accident was clearly 'manmade' resulting from collusion between the government, the regulators and the Tokyo Electric Power Company (TEPCO), and the lack of governance of these parties (The National Diet of Japan, 2012, p. 16). Public trust was also eroded by the scandal of a number of post-Fukushima public hearings which took place before the 2012 national DP (Sieg, 2012). Our primary qualitative data also suggest that pre-existing distrust seems to be highly relevant to participants' lack of trust in the government's motive in organising this DP with some participants raised concerns that the government may "make stories and scripts" and manipulate the DP for its own sakes (Illustrative Examples No. 20, 21, 23, 24).

In terms of the political context, our empirical findings indicate that the links between trust and policy legitimacy can be influenced by the political situations of energy policy-making systems. It was evident that two political developments re-inforced public distrust in the motives of the government in organising the DP. The first was the imminent regime shift from the DPJ to the LDP government at the time when the DP was conducted. The second was the prevalence of the 'nuclear village' – a pro-nuclear coalition of bureaucrats and the nuclear industry – which has continued to exercise significant political influence to lobby against the phasing-out of nuclear power even after the Fukushima nuclear accident (Kingston, 2014; Suzuki, 2019). The polarised debate between the "pro" and "anti" sides of the nuclear power issue in Japan has remained in the last several years, indicating the lack of policy legitimacy in nuclear decisions made by the Japanese government (Suzuki, 2019). The participants' perception of a lack of political commitment (Illustrative Examples No. 32 and 33) and a lack of perceived policy impacts (Illustrative Examples No. 34-40) appeared to undermine their trust in the motive of the government in conducting the DP.

6. Conclusion

Public trust and the resulting legitimacy of energy transitions policy can be affected by how deliberative policy-making processes take place. However, deliberative policy-making on its own cannot guarantee trust and policy legitimacy. By testing our framework in the case of the 2012 national DP in Japan, this study contributes to the debate on the potential and limits of deliberative policy-making by conceptualising the trust gap – which is a critical element of deliberative processes that affect the outputs.

Our study advances the literature on deliberative policy-making in energy transitions in three important ways. Firstly, we have demonstrated how our trust-based systems approach can be used to highlight the complexity of the deliberative processes. We have showed that the trust gap is an important element of deliberative policy-making processes. We particularly advance the literature by providing a more nuanced explanation of the complexity of the trust gap by specifying the three critical dimensions of distrust in information, distrust in motive, and distrust in competence. We have also specified the three directional dynamics of the trust gap: vertical, horizontal, and temporal. In addition, we identified two key contextual factors (the distrust context and political environments) which appeared to inhibit closure of the trust gap even though deliberative processes provided an opportunity to do so. Although the 2012 national DP cannot be regarded as a successful experience in terms of trust-building, our conceptualisation of the trust gap makes a contribution to the literature on the dimensionality and dynamism of trust in deliberative policy-making (Fenenga et al., 2015). Our case study confirms with the literature that legitimacy is associated with trust in governing bodies, which can be undermined by the absence of demonstrated political commitments and perceived effectiveness in producing outcomes; these are

important indicators of the motive dimension of trust (Turner et al., 2016). These findings confirm earlier work that the normative outcomes of deliberative participatory approaches are politically constrained (Safaei, 2015) and that deliberative policy-making is a political process in which citizen recommendations compete with claims by interest (stakeholder) groups and elected or appointed decision-makers (Renn et al., 1993).

Secondly, our study has advanced the systems theory literature by drawing attention to the process of deliberative policy-making systems for energy transitions, and bridging the input-output gap (Corbett et al., 2018; Schmidt, 2013). Our trust-based systems framework is process-oriented, arguing that we need to look beyond outputs to focus on the deliberative processes from a trust perspective. Our findings not only confirm with the literature that legitimacy is not an automatic output of a deliberative policy-making process, given that legitimacy involves highly complex dynamics including public (dis)trust, but also showing the importance of the distrust context and political context in constraining the outputs of deliberative policy-making processes.

Thirdly, this study contributes to the literature by testing our framework using a case study of the 2012 national DP in Japan. Existing work has not yet provided sufficient empirical support for the proposition that there should be a normative and positive link between deliberative approaches and policy outputs in the context of energy transitions, especially in the context beyond the west. Compared with western developed countries, Asian countries are different in economy, politics, natural endowments, and culture (AASA, 2011), which may lead to variations of energy transitions pathways, processes as well as outcomes. This study has traced the deliberative process of the 2012 national DP. We have shown that a normative trust-building process did not take place, and the ideal of policy legitimacy was undermined by complex contextual factors associated with public distrust and politics. Our case study thus partially fill in the gap by demonstrating that the concepts of trust and deliberative policy-making which originated in West are relevant in Asia.

The findings of this study also have policy implications, particularly in relation to how governments can proactively develop trust-building deliberative strategies. Firstly, policy-makers need to give sufficient attention to trust when designing deliberative participatory strategies. Our findings suggest that trust should be a strong component of the deliberative policy-making process. Failure to adequately consider the trust perspective may undermine the legitimacy of policies intended to support energy transitions. Our findings suggest that three aspects of deliberative and participatory events and processes should be the focus in order to gain trust: recruitment methods, information provision, and scenario development methods. This study has shown that these three elements were particularly critical in affecting participants' perception of trust in the 2012 national DP. More inclusive, transparent, and open approaches to designing a deliberative exercise are thus desirable in order to accrue public trust.

Secondly, the political issues identified in the case study suggest that trust cannot be built only through deliberative exercises such as DPs or other forms of participatory mechanisms. Much more attention needs to be given to the broader context of public distrust and political situations. Policy-makers should focus their attention on the institutionalisation of participatory mechanisms in order to enhance policy consistency even across shifts in political regimes (McLellan et al., 2017). The 2012 national DP showed that a change in political regime can lead to a major shift from an anti- to a pro-nuclear policy in a matter of months, with the citizens' considered views on 0% nuclear being disregarded as a result. Some good examples of institutionalisation as a means of trust-building can be found in France, which has introduced the Transparency and Security Act (TSN) (2006) and the Energy Transition for Green Growth Act (2016) to safeguard the public's right to access reliable information on nuclear matters through institutionalisation (IEA, 2016; NSA, 2006).

This paper is based on a single case study of the 2012 national DP, the first and a rare attempt at deliberative participation in energy policy-making at a national scale in Japan. This was followed by another major national nuclear DP in South Korea in 2017 (Public Deliberation Committee, 2017). Study of the 2012 national DP as a precedent is therefore of academic importance, offering a crucial context for comparative studies in this field. Comparative studies across countries (within Asia, such as South Korea and Japan, and between Western and Asian countries) and energy options (for example, nuclear and wind power) would be a fruitful avenue for future research. Such comparative studies will ultimately provide insights into the sociopolitical aspects of energy transitions.

Declaration of Competing Interest

The authors report no declarations of interest.

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Appendix A. Pre- and post-forum questionnaires.

The same set of questions were used in the pre- and post-forum questionnaires to check for changes in attitudes. Source: CDD (2012c).

Q1-1. We will ask you about some important aspects when choosing energy sources and method for their generation.
How important are the following factors to you from 1 to 4? On a scale from 0 to 10, where 0 is not important at all, 10 is the most important, and 5 is exactly in the middle, please choose one for each question.

1 Safety
2 Stable supply
3 Prevention of global warming
4 Cost

Q1-2. In terms of the choice of energy source and generation method, which of the four above aspects is most important to you – safety, stable supply, prevention of global warming, or cost? I will read the aspects so please rank the top three in order of importance from 1st to 3rd.

A. Which is the most important?
1 Safety
2 Stable supply
3 Prevention of global warming
4 Cost
5 Don't know

B. Which is the second most important?
1 Safety
2 Stable supply
3 Prevention of global warming
4 Cost
5 Don't know

C. Which is the third most important?
1 Safety
2 Stable supply
3 Prevention of global warming
4 Cost
5 Don't know

Q2. We will ask you about nuclear power generation in 2030, which is 18 years from now. Do you agree or disagree with the statements from A to C below? On a scale from 0 to 10, where 0 is strongly disagree, 10 is strongly agree, and 5 is exactly in the middle, please choose one for each question.

A Do you agree with the statement "We should abandon all nuclear power plants as soon as possible, shutting down all plants by 2030"? On a scale from 0 to 10, where 0 is strongly disagree, 10 is strongly agree, and 5 is exactly in the middle, please choose one.
B Do you agree with the statement "We should reduce nuclear power generation gradually (as a result, the ratio of nuclear power generation in total electric power will be about 15 %)"? On a scale from 0 to 10, where 0 is strongly disagree, 10 is strongly agree, and 5 is exactly in the middle, please choose one.
C Do you agree with the statement "We should keep nuclear power generation on a lower ratio as compared to the current level (as a result, the ratio of nuclear power generation in total electric power will be about 20−25% in 2030"? On a scale from 0 to 10, where 0 is strongly disagree, 10 is strongly agree, and 5 is exactly in the middle, please choose one.

Q3. Now we will ask you about nuclear power generation and energy policies in the following sections A to I.

A Some people believe that "We should not use nuclear power generation because it is difficult to ensure its safety." Other people believe that "We should keep nuclear power generation because we can ensure the safety of nuclear power generation." On a scale from 0 to 10, where 0 is "we should not use", 10 is "we should keep using", and 5 is exactly in the middle, please choose one.
B Some people believe that "In order to prevent global warming, we should promote the use of renewable energy and energy conservation even if costs will increase." On a scale from 0 to 10, where 0 is strongly disagree, 10 is strongly agree, and 5 is exactly in the middle, please choose one.
C Some people believe that "We should change our life-style by reducing energy and electric power consumption significantly even if it means our lives will become less convenient in comparison to how we live now." On a scale from 0 to 10, where 0 is strongly disagree, 10 is strongly agree, and 5 is exactly in the middle, please choose one.
D Do you agree with the statement "Because the final spent nuclear fuel affects future generations, we have to quit using nuclear power as soon as possible"? On a scale from 0 to 10, where 0 is strongly disagree, 10 is strongly agree, and 5 is exactly in the middle, please choose one.
E Some people believe that "The Japanese people have already saved enough energy.." Other people believe that "The Japanese people should save more energy.." On a scale from 0 to 10, where 0 is saved enough, 10 is should save more, and 5 is exactly in the middle, please choose one that reflects your opinion.
F Some people believe that "Japanese industries have already saved enough energy.." Other people believe that "The Japanese industries should save more energ.." On a scale from 0 to 10, where 0 is saved enough, 10 is should save more, and 5 is exactly in the middle, please choose one that reflects your opinion.
G Some people believe that "the government has promoted renewable energy such as solar and wind power enough." Other people believe that "the government should do their best to promote renewable energy." On a scale from 0 to 10, where 0 is "promoted enough", 10 is "should do their best", and 5 is exactly in the middle, please choose one that reflects your opinion.

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(*continued*)

H Some people believe that "we have to solve the energy issue as a community so we have to adopt the policy "buy and use locally." Other people believe that "It is efficient to create a large volume of electricity as a nation." On a scale from 0 to 10, where 0 is "should solve locally", 10 is "should solve nationally", and 5 is exactly in the middle, please choose one that reflects your opinion.

I Some people believe that "Switching energy policies should be done by the government responsively." Other people believe that "Each individual needs to ready themselves for a switch in energy policy and change their thinking." On a scale from 0 to 10, where 0 is "government", 10 is "individuals", and 5 is exactly in the middle, please choose one.

Q4. In order to generate enough energy around 2030, which energy we should use? On a scale from 0 to 10, where 0 is "should not use at all", 10 is "should use as much as possible", and 5 is exactly in the middle, please choose one that reflects your opinion.

a Oil
b Coal
c Natural gas
d Hydro
e Nuclear
f Solar
g Wind
h Biomass
i Cogeneration
j Geothermal

Q5. For the following questions A to D, how much do you feel you can trust each source? On a scale from 0 to 10, where 0 is do not trust at all, 10 is trust very much, and 5 is exactly in the middle, please choose one for each question.

a Information from the government
b Information from nuclear power specialists
c Information from electric companies
d Information from media
e Information from NPO and NGO
f Information from the Internet

Q6. Do you think the following ideas are important personally and for our society? On a scale from 0 to 10, where 0 is the least important, 10 is the most important, and 5 is exactly in the middle, please choose one.

a All people should have equal opportunities.
b It is ideal for people and companies to compete freely with each other.
c We should minimize our unemployment rate.
d No one should struggle with food or having a place to live.
e We should earn as much money as possible.
f We should have a choice on how to use our own money.
g We should minimize gaps in income and assets.
h We have to promote economic growth.
i Children should receive good educations.
j We should have a safe community.
k All people should have clean air and water.

Q7. Do you agree or disagree with the following statements? Please choose from "disagree"; "neither"; "agree"; "no opinion/NA".

a Politicians are concerned about what ordinary people are thinking.
b Because most public policies are complicated, it is impossible for ordinary people like me to understand them.
c It makes no difference even if ordinary people like me say something against what the government does.
d I have opinions and want to ask someone to listen.

Q8. Please imagine people who have different opinions from yours. Do you agree or disagree the following statement? Please choose from "disagree"; "neither"; "agree"; "no opinion/NA".

a They do not understand the issues.
b They believe things that are actually not true.
c They did not think about (the issues) clearly.
d They have some reasons. These reasons in some aspect are solid.
e They only pay attention to their own benefits.

Q9. Please imagine people who have different opinions from yours. Do you agree or disagree the following statement? Please choose from "disagree"; "neither"; "agree"; "no opinion/NA".

a I respect other people's opinion even if they are different from mine.
b It is impossible to reach agreements with them.
c I can compromise for the purpose of achieving agreements that both sides of people can support.

Appendix B. An overview of the changes in political regimes, nuclear strategies, and nuclear policies in Japan between 2010 and 2018.

Sources: (Kim and Chung, 2018; White, 2015; WNA, 2019).

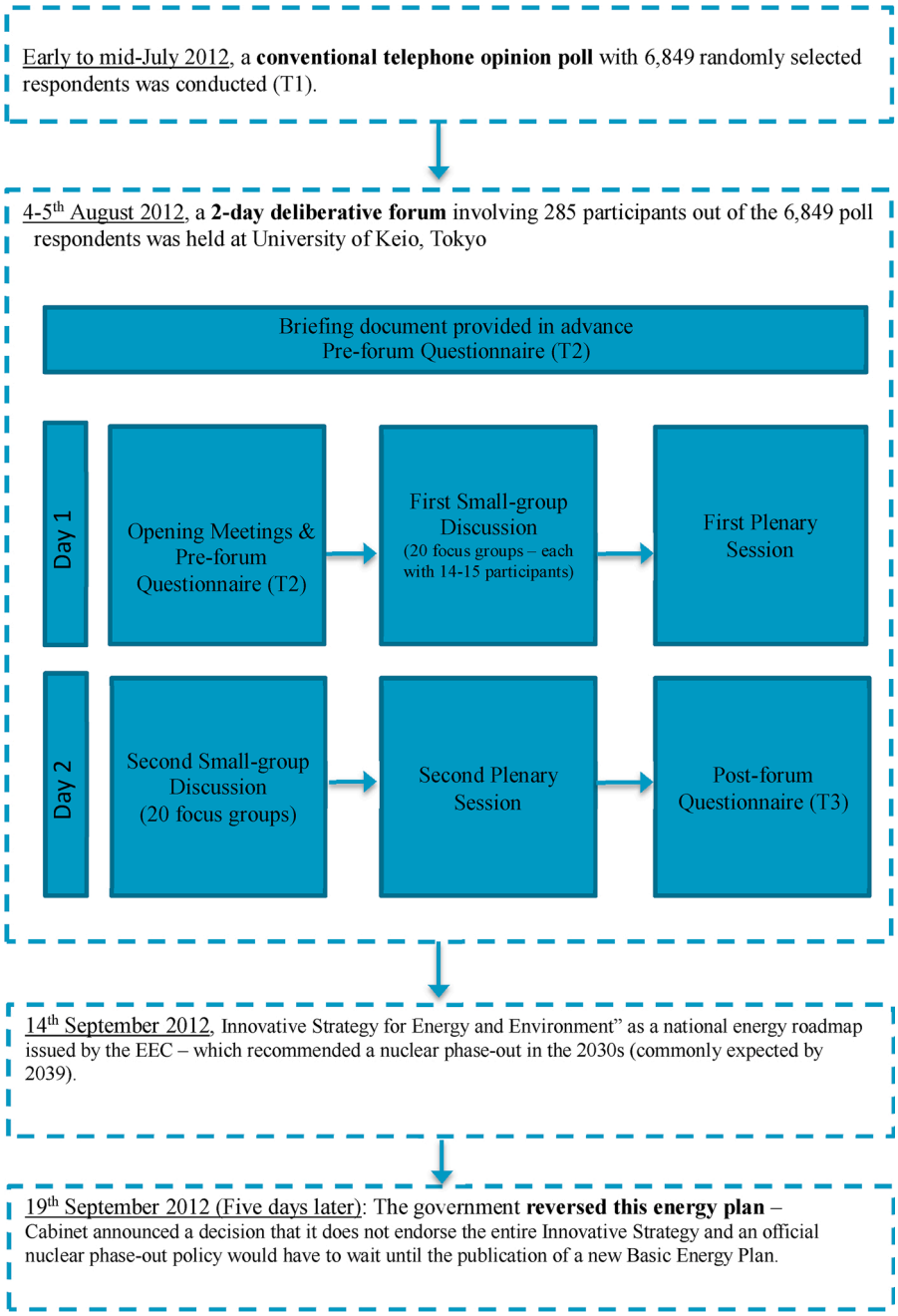
	Government/ Political Party	Nuclear Strategy	Major Nuclear Policies	
Pre-Fukushima	Kan Administration (8/6/2010-2/9/2011) DPJ	Ambitious nuclear expansion development	- In 2010, 25 % of electricity from nuclear power - In September 2010, • Emphasized on 3Es: energy security, environmental protection, and efficient supply • Set 50 % of electricity from nuclear power in 2030 • 14 more reactors planned to be built by 2030	
		Nuclear plan on hold, and reconsidered	7/2011	EEC was established to recommend national energy strategies
			7/2011	Most reactors were shut down
	Noda Administration (2/9/2011-26/12/2012) DPJ	Nuclear phase-out	10/2011	The White Paper on Energy Policy was published, confirming a nuclear reduction plan
			4-6/2012	A period of no nuclear
			7-8/2012	The 2012 national DP: A period of public engagement for deciding nuclear power policy
			9/2012	Nuclear phase-out in the 2030s (commonly expected by 2039) was recommended but not officially endorsed as a policy: • 6 September 2012, DPJ published a party policy report that states the government will enable 0% nuclear in the 2030s • 14 September 2012, the EEC issued the "Innovative Strategy for Energy and Environment (Innovative Strategy)" which recommended a nuclear phase-out in the 2030s (commonly expected by 2039). • 19 September 2012, the Cabinet Decision was announced which did not endorse the Innovative Strategy entirely. An official nuclear phase-out policy would have to wait for the publication of a new Basic Energy Plan The Nuclear Regulation Authority (a new nuclear regulator) was established
Post-Fukushima	Abe Administration (26/12/2012 - present) LDP	Nuclear restart	12/2012	LDP won the General Election
			7/2013	Stricter regulations for nuclear reactors were finalised; nuclear restart applications have begun to be reviewed
			9/2013	Oi reactors were shut down for reviewing, which marked the start of the second period of no nuclear
			4/2014	New 4th Basic Energy Plan presented: • Nuclear power regarded as "important power source that supports the stability of the energy supply and demand structure" • Degree of dependence on nuclear should be reduced
			6/2015	Plan for Electricity Generation to 2030 approved: • Nuclear at 20−22% in 2030 • Renewables 22−24% • LNG 27 % • Coal 26 %
			8/2015, 10/2015	Two reactors in Sendai, Miyagi Prefecture were restarted
			2/2016, 3/2016	Two reactors in Takahama, Fukui Prefecture were restarted. But a subsequent district court injunction forced the two reactors to shut down
			4/2016	Electricity market deregulation expanded to the retail segment
			4/2017, 5/2017	Two reactors in Takahama, Fukui Prefecture were restarted following the lift of court injunction
			7/2017	Draft Basic Concept on Nuclear Energy Use approved, providing reference for future decisions about nuclear energy policy
			3/2018, 5/2018	Two reactors in Ohi, Fukui Prefecture were restarted
			3/2018, 6/2018	Two reactors in Genkai, Saga Prefecture were restarted
			7/2018	The 5th Basic Energy Plan approved: • Electricity generation plan agreed in mid-2015 maintained in the latest plan • Nuclear power regarded as "an important base-load power source contributing to the stability of the long-term energy supply-and-demand structure"
			11/2018	One reactor in Ikata, Ehime Prefecture was restarted

Appendix C. Major participatory policy-making initiatives/exercises in recent years in Japan.

Types	Time	Policy Areas	Description	Sources
Public Hearings	Mar 1994	Nuclear Energy	• First-ever public hearing on Japan's nuclear energy policy • Organized by the Japan Atomic Energy Commission (JAEC)	(White, 2014)
Round Table Conferences	1996, 1998, 1999	Nuclear Energy	• A sodium accident occurred at the Monju prototype fast breeder reactor • Government set up a series of round table conferences on this issue	(White, 2014)
Referendums	1996	Nuclear Energy	• First local referendum in Maki • About the construction project of nuclear power plant	(White, 2014)
	1997	Nuclear Energy	• Second nuclear-related referendum in Kariwa • Related to the implementation of the pluthermal program	
Consensus Conferences	2000	Food Safety	• A national-level consensus conference • About genetically modified agricultural products	(White, 2014)
	2001, 2003	Food Safety	• The same topic consensus conference nationwide • Hokkaido specified on genetically modified crops	
Deliberative Polls	Dec 2009	Administrative Division	• DP conducted in Kanagawa • A public opinion poll on Doshu-system (Federated Regional System)	(Authors)
	Jan 2010	City Planning	• "One Day Deliberation" held in Fujisawa • To help make a new Comprehensive City plan	(CDD, 2019)
	Aug 2010	City Planning	• A second DP conducted in Fujisawa • To help make a new Comprehensive City plan	(Authors)
	May 2011	Welfare System	• "Pension System, A Generation Choice" DP held nationwide • To collect opinions about pension system in Japan	(CDD, 2019)
	Nov 2011	Food Safety	• "Mad Cow Disease" DP held in Sapporo	(CDD, 2019)
	Aug 2012	Energy and Environment	• DP on energy option and dependence on nuclear power in 2030 (focus of this paper)	(Sone, 2014)
	Mar 2014	Weather and Environment	• DP about snow measures in Sapporo	(Authors)
Planning Cells	By end of 2010	Various Issues	• Many planning cells had been held	(White, 2014)
Town Meetings	FY2010	Energy	• "Town Meeting on Environment and Energy" held nationwide • In Tokyo, Sapporo, Osaka, Nagoya, Sendai, Toyama, Fukuoka, Hiroshima, Saitama, Takamatsu and Okinawa	(METI, 2010)

Appendix D. A timeline of the 2012 national DP and the policy impacts.

Source: Authors; data from CDD (2012a); Transcript: A002.



Appendix E. An assessment of trust in the deliberative forum of the 2012 national DP.

Dimensions of Trust	Trust-related indicators of deliberative processes	Qualitative data – Illustrative examples
Trust in Information	Transparency	1 This poll is using the RDD (Random Digit Dialing) to call out participants who are willing to join, hence I would like the organiser to publish the outline of the poll including the participant list. In other words, my first point is that target group of the poll, the poll organiser, the stratification of sample, like gender, age and district, should be *openly informed*. The result comes from only people who are caring about the issue recruited by the poll, at this point I think it reveals the limitation of the poll already. (Transcript B007)
	Comprehensiveness	2 We talked about the booklet yesterday. The writers wrote this book according to the *information from the government only*. Therefore, I think something in *the information is lacking*. (Transcript B009) 3 Therefore, just like what you guys said, we know nothing because the *government* and the *media* just give the numbers to us without details. (Transcript B002) 4 I was surprised that the proposal given by the government is without details. (Transcript B014) 5 I find it strange that thing explaining the cause for the accidents is not written. (Transcript B017) 6 Will the nuclear plants be closed today or tomorrow? Will some of them restarted? There is *nothing about the process written here*. (Transcript B018) 7 Our group *wants to know more about the benefits and downsides* of the 0% nuclear option. We are particularly interested in knowing more about the negative impacts on the economy. (Transcript A002) 8 The 311 Fukushima nuclear accident has brought to the collapse of energy security. The term "energy security" is all over the place in the information that I received yesterday. I therefore hope that the assessment criteria of energy security *can be elaborated in a more solid manner*. Although I appreciate that it could be difficult, please use data to explain the term "energy security". (Transcript A002) 9 Actually, we don't know how much we need to spend in order to make clean energy. It is just like a dream. As there are *no figures and details about the clean energy, we cannot proceed further in our discussion.* (Transcript A002)
	Objectivity and reliability	10 In page 9, there is something called "The first prediction on domestic energy supply". If we look at the water power in it, the supply of electricity from water power remained the same from 1965 to 2009. The number written in blue margin doesn't change. Take a look on the last year, 2009, it is 3.2%. I think that it is impossible for this number. From the previous graphs, it is 10%. *Isn't the number entirely groundless*? (Transcript A003) 11 We talked about fossil fuels. I am nearly 40 years old. In the information, it is written that the oil will not be oil in 40 years. When I was small, I was also told that the oil would run out in 40 years. So, how many 40 years are that in fact? *Is the information correct?* It is a must for us to express our opinions but the information should be correct. (Transcript A008) 12 However, *we don't know whether the information given is correct.* Then, it seems that the information is just different from what the people with relevant knowledge said. (Transcript A009)
Trust in Motives	Openness	13 There are three *given scenarios. I just feel that the government wants to keep nuclear power at 15%. I don't trust the government* and I don't think that this plan is feasible. (Transcript B002) 14 I think that we are just *misled to choose the 15% scenario* because we are told that we could take the middle one among the three choices. (Transcript B003) 15 So, we will choose 15 %. But *why the middle option must be 15%? Can't it be 10 or 5%? I feel that we can just take the middle option.* Because of forcing us to choose 15% scenario, the government gave us the 25% scenario. That's dirty. (Transcript B004) 16 First of all, I want to ask for the reason *why we have to jump from zero to 15%.* Can't we have any options between 1−10% or 11−15%? …I just feel that *we are misled to choose 15%* nuclear power because it is the option between zero and 20−25% nuclear power scenario…. Yesterday, we decided the numbers zero, 15% and 20−25% but there was kind of answer, saying that we should have 15% nuclear power after 2030. If we choose zero, it will be zero forever. If we choose 20−25%, we will have 20−25% nuclear power forever. Isn't 15% a vague number? It is just like a pitfall. …Because of the questionnaires we did, the conclusion of the report will say that we support 15% scenario, which is not really our will…. Why they number jumps from 0 to 15? There must be reasons opposing 15% scenario. I think that we should add more scenarios in the space between 0 and 15 like 5,6,7 or 8%. (Transcript B011) 17 That's a bit naive. I feel that *we are forced to make this choice not to choose the 0*% scenario (Transcript B014). 18 *I personally would want to oppose absolutely, I have a weird feeling that I am made to choose between* 25 % *or* 15 %. (Transcript B017). 19 Nuclear is a cruel energy option, there is a consensus that "if possible, we do not want to use it". But why is "zero-nuclear by 2030" included as an option? We are clear that there are negative impacts brought by using nuclear energy, but now we are put into a situation to think that it is not feasible to implement zero-nuclear". Why other choices were not included as scenario options? (Transcript A002).

(*continued on next page*)

(*continued*)

Dimensions of Trust	Trust-related indicators of deliberative processes	Qualitative data – Illustrative examples
Trust in Motives	Integrity	20 I have heard that people related to *power companies were blend into the crowd in a meeting*. So, when I looked at this once again, I started to doubt whether this is really ok. (Transcript A006) 21 *The government officials will just make stories and scripts*…The government only wants to maintain the level of nuclear power… No I don't think so. It is just a game. They will just choose the options for their own benefits. (Transcript B011) 22 Can they make a good conclusion? It seems that the *government just want to create the fact that they did listen to the public but not really accepting the public opinions*. (Transcript B011) 23 Isn't the deliberative forum only organised *as a gesture of the government*? (Transcript B017) 24 I can only think that the *government is pulling strings behind the whole thing*. (Transcript B017) 25 I understand that the government needs to investigate whether the public are holding the same opinion but why the government need to know how the opinion changes*? I care about the way they use the opinions* of the citizens. (Transcript A015) 26 Although it is said that the government would refer to the results of the discussion and make a decision in August, I want to ask how much our opinions could be reflected and where is the protection for us. (Transcript A019)
	Inclusiveness	27 I think that we are (mis)led. After reading the document I think that *the government may probably let the people of opposite stances to quarrel and take the middle thing*. (Transcript A017) 28 Some kinds of renewable energy are recorded in the booklet [the briefing document] like solar power but there is no introduction to marine power and wave power. Does *the government refuse to accept* these kinds of energy? (Transcript A018)
	Representativeness	29 As I was allowed to join, I just thought that I could come and oppose the nuclear power and *I thought that the people who support nuclear power will not come*. Therefore, there may be one to two people supporting nuclear power. As a result, the opinions regarding nuclear power may be one sided and no people will support nuclear power anymore. So I think that we'd better have some people who support the nuclear power. (Transcript B003) 30 Most of the people there oppose to nuclear power. Is it a bit biased? Are we coming here for a fair discussion? (Transcript B011) 31 In my group, there are many participants who supported zero-nuclear. But is it desirable to phase-out nuclear technology? (Transcript A002)
	Credibility in political commitments	32 *Does 'government' mean the current Democratic Party of Japan?* They have been talking *about, in extreme situations that they might disband in August*. If this is true, does what says here reflect the reality? Is it the government raised by the Democrats, or the bureaucracies behind? Even the government has changed, this energy problem would continue, it is good to divide the framework into 0, 15, 25? And the question before all these is what is 'government'? (Transcript A006) 33 Three nuclear options of 0%, 15 % and 30 % are given to us to make a choice. But Japanese citizens have distrust. *When we think about the possibility of political regime changes, we feel uncomfortable*. (Transcript A002)
	Perceived policy impacts	34 I have an impression that even we made a decision, the *government will not follow as the government is the final decision-maker*. (Transcript B005) 35 As the government didn't explain whether nuclear power is safe in details, it is not able to make a decision. *I don't know what's the meaning for the discussion*. (Transcript B015) 36 As we said yesterday, Mr. Hosono*, the minister of Environment was looking at the panelists behind the wall, right*? When the experts were answer the questions, he was laugh at the back stage. I want to ask whether the government has a stance for the issue. Do they agree or disagree? *I noticed that the government has already made a decision*. (Transcript B013) 37 As a result I question about the purpose of the forum. Is it held for letting the government know our opinions? Even so*, the government roughly listened our opinions but the government has already had a plan*. It seems that *we are discussing without really affecting the policy making*. (Transcript B003) 38 I question about *whether our opinions will be delivered to the government and whether the government will follow our suggestions?* (Transcript A004) 39 I question about *how much our opinions will be reflected*. (Transcript B009) 40 What will become of the questions we raised? Where will it go? *Where will the results of this conference be?* Although you said it is because you want to get the details of the change reflected in the questionnaire, but is this topic suitable? (Transcript B017)
Trust in Competence	Participants' perception of the government's capabilities in implementing energy alternatives	41 What is the timeline of implementing these energy plans? Do experts have in hand the required data? We do have serious doubt on it. (Transcript A002) 42 I have great doubt about whether the real detailed plan could be done by end of August. Even for us it is only this level. When we were told to come during the questionnaire, "oh, that sounds interesting, let's go" is our true voice, and the result is more people who relatively care about this issue had come than those people who don't care at all. And the discussion made by such people is only at this level. And can such conclusion made here become the summary of what will be done by August? (Transcript B008)
	Participants' perception of the government's competence in managing risks	43 I *feel very anxious after looking at the reaction of the government* and TEPCO after this nuclear accident. Even they told me it is fine now, I do not trust them. (Transcript A006) 44 I *used to believe* in the government before the nuclear accident *but I changed my mind* after the accident. (Transcript B012) 45 The type of nuclear power, which is not burning, is not feasible due to the high cost. Why we still spend money on it? Why don't we put more money of this and investigate more on it? *The act of the government is slow*. (Transcript A015)

Notes:

(1) *Italics*: emphasis added by authors
(2) The transcripts are coded as follows:

Code A001 – Transcript of the plenary session on the first day of the deliberative forum.
Code A002 – Transcript of the plenary session on the second day of the deliberative forum.
Other codes: Small-group discussion sessions.

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